PRACTICAL VISUALIZATION OF CYBERSECURITY DATA USING GENERAL-PURPOSE AND CYBER-SECURITY TOOLS

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The number and complexity of cybersecurity threats have significantly expanded due to the quick development of digital systems and connectivity. Attacks against organizations can take many different forms, such as ransomware, malware, phishing, DDoS, and advanced persistent threats (APTs). It takes more than just raw log data to comprehend, communicate, and address these dangers effectively; it also requires a clear visual depiction of patterns, trends, correlations, and anomalies.

A key component of contemporary cybersecurity operations is data visualization. Security analysts can use it to swiftly spot anomalous activity, spot trends of attacks over time, comprehend the connections across infrastructure, track the development of campaigns, and share findings with both technical and non-technical stakeholders. While cyber-specific tools like Maltego, Kibana, or Splunk are excellent at link analysis, threat hunting, and relationship mapping, general-purpose visualization libraries like those in Python enable flexible statistical and exploratory research.

This assignment examines both kinds of technologies to show how visualization aids in threat comprehension and decision-making, as well as to display cybersecurity-related data.

1.2 Aim of the Assignment

Applying visualization techniques to cybersecurity-related datasets using one general-purpose tool and one cyber-specific tool, as well as clearly documenting and explaining the generated visuals and the insights they offer, are the objectives of this practical project.

1.3 Objectives of the Assignment

The main objectives are:

- To select and use one general-purpose visualization tool (Python with Matplotlib and Seaborn) to create three different types of diagrams from a cybersecurity dataset.

- To select and use one cyber-specific visualization tool (Maltego Community Edition) to produce at least three diagrams demonstrating different cybersecurity analysis purposes.
- To clearly document the datasets used, the justification for tool selection, the step-by-step creation process, and the meaning and insights gained from each visualization.
- To demonstrate the entire workflow through a video tutorial and a well-structured written report.
- To strictly use only public, sanitized, or simulated datasets and to avoid any live attack data or illegal sources.

1.4 Significance of the Assignment

Effective visualization is essential in cybersecurity for several reasons:

- It transforms complex, high-volume data into understandable patterns and anomalies.
- It supports faster detection, better communication between analysts and decision-makers, and more effective incident response.
- It bridges the gap between raw technical data (logs, traffic, IOCs) and human interpretation.
- It helps demonstrate the complementary strengths of general-purpose data science tools and specialized threat intelligence platforms.

This task aids in the development of real-world abilities that are directly relevant to incident response, threat intelligence teams, security operations centers (SOCs), and cybersecurity academic research.

1.5 Scope and Limitations

This work is limited to:

- Visualization only – no machine learning model training, no statistical hypothesis testing, no real-time monitoring.
- Use of exactly two tools: Python (Matplotlib + Seaborn) and Maltego Community Edition.

- Creation of three visualizations per task (total six diagrams).
- Use of only publicly available, sanitized, or synthetic datasets (no live attack data, no private or illegal sources).
- Focus on explanatory documentation and video demonstration rather than production-level dashboard deployment.

The assignment does not cover advanced deployment (cloud dashboards, real-time feeds) or integration with SIEM systems.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of Visualization in Cybersecurity

Data visualization is now a crucial part of contemporary cybersecurity procedures. As the amount, speed, and diversity of security-related data (network traffic, logs, threat intelligence feeds, phishing reports, endpoint telemetry, etc.) increase, analysts increasingly depend on visual aids to spot trends, recognize abnormalities, comprehend attack campaigns, and effectively convey findings.

Visualization helps bridge the gap between raw data and human cognition, according to numerous research and industry reports (e.g., SANS Institute, MITRE ATT&CK framework documentation, and academic publications on security visualization). It facilitates quicker threat identification, helps with incident triage, facilitates threat hunting, and enhances management-technical team communication.

2.2 General-Purpose Visualization Tools in Cybersecurity

Cybersecurity professionals frequently use general-purpose visualization tools like Matplotlib, Seaborn, Plotly (Python), ggplot2 (R), Tableau, and Power BI for exploratory data analysis and reporting.

Literature highlights several advantages of Python-based tools (Matplotlib and Seaborn) in security contexts:

- High flexibility and customizability
- Ability to handle large datasets through integration with pandas and NumPy
- Strong support for statistical visualizations (heatmaps, box plots, violin plots, scatter plots with hue)
- Open-source nature and reproducibility
- Integration with Jupyter notebooks for interactive analysis

Common use cases include:

- Temporal trend analysis of attacks (line charts)
- Frequency distribution of attack types (bar/pie charts)
- Correlation and outlier detection (scatter plots)
- Pattern discovery in time-of-day or geo-location data (heatmaps)

2.3 Cyber-Specific Visualization Tools

Particularly in the areas of link analysis, entity resolution, and relationship mapping, special-ized tools made for cybersecurity and threat intelligence offer features that general-purpose tools do not deliver as well.

Maltego is one of the most widely referenced tools in the literature for open-source intelligence (OSINT) and threat infrastructure visualization. Key strengths include:

- Entity-based graph modeling (domains, IPs, emails, persons, ASNs, BTC addresses, etc.)
- Automated transforms that pull data from public sources (DNS, WHOIS, Shodan, VirusTotal, etc.)
- Support for threat campaign mapping, phishing infrastructure analysis, and attack path reconstruction
- Visual relationship discovery that reveals hidden connections

Other commonly mentioned cyber-specific tools in recent literature include:

- Kibana / Elastic Stack (for log and time-series visualization)
- Splunk (dashboards and search-driven analytics)
- Grafana + Loki / Prometheus (observability and security monitoring)

2.4 Visualization Types in Cybersecurity Literature

Recent papers and practitioner guides commonly discuss the following visualization types in cybersecurity contexts:

- **Bar and Pie Charts** – used for categorical frequency analysis (attack types, severity levels, protocol distribution)
- **Line and Area Charts** – temporal trend analysis (attack volume over time, campaign evolution)
- **Heatmaps** – two-dimensional pattern discovery (time-of-day × weekday, source/destination IP relationships, port usage)
- **Scatter Plots** – relationship and outlier analysis (packet size vs anomaly score, latency vs threat score)
- **Graph/Network Diagrams** – entity relationship mapping, infrastructure graphs, kill-chain visualization
- **Timelines** – event sequencing and campaign reconstruction

2.5 Importance of Appropriate Visualization Selection

Multiple authors emphasize that choosing the correct visualization type is critical:

- Wrong chart type can obscure important patterns or mislead interpretation
- Effective visualization must match the analysis question (comparison, distribution, trend, relationship, composition)
- Context (audience, interactivity needs, data volume) must be considered

2.6 Summary of Literature Review

The literature study emphasizes how important visualization is to cybersecurity analysis and decision-making. For exploratory and statistical visualization tasks, general purpose tools particularly Python-based libraries offer accessibility, statistical depth, and flexibility. Relationship discovery, entity linking, and threat infrastructure mapping are skills that are hard to accomplish well with general-purpose tools alone, but cyber-specific tools (most notably Maltego) excel at them.

By practically implementing one general-purpose tool (Python with Matplotlib/Seaborn) and one cyber-specific tool (Maltego) to actual cybersecurity datasets, this assignment expands upon these foundations and demonstrates both methodologies in a repeatable and thoroughly documented way.

CHAPTER 3

IMPLEMENTATION AND RESULTS

This chapter presents the practical work carried out for both tasks of the assignment. It includes:

- Environment setup and tool installation
- Dataset loading and basic exploration
- Step-by-step creation of the required visualizations/diagrams
- Screenshots of all major steps and final results
- Brief explanation of each visualization

3.1 Task 1 – General-Purpose Visualization Tool (Python)

3.1.1 Tool and Environment Setup

The general-purpose tool selected is Python with the libraries `pandas`, `matplotlib` and `seaborn`. All work was performed in a virtual environment using Jupyter Notebook on Ubuntu.

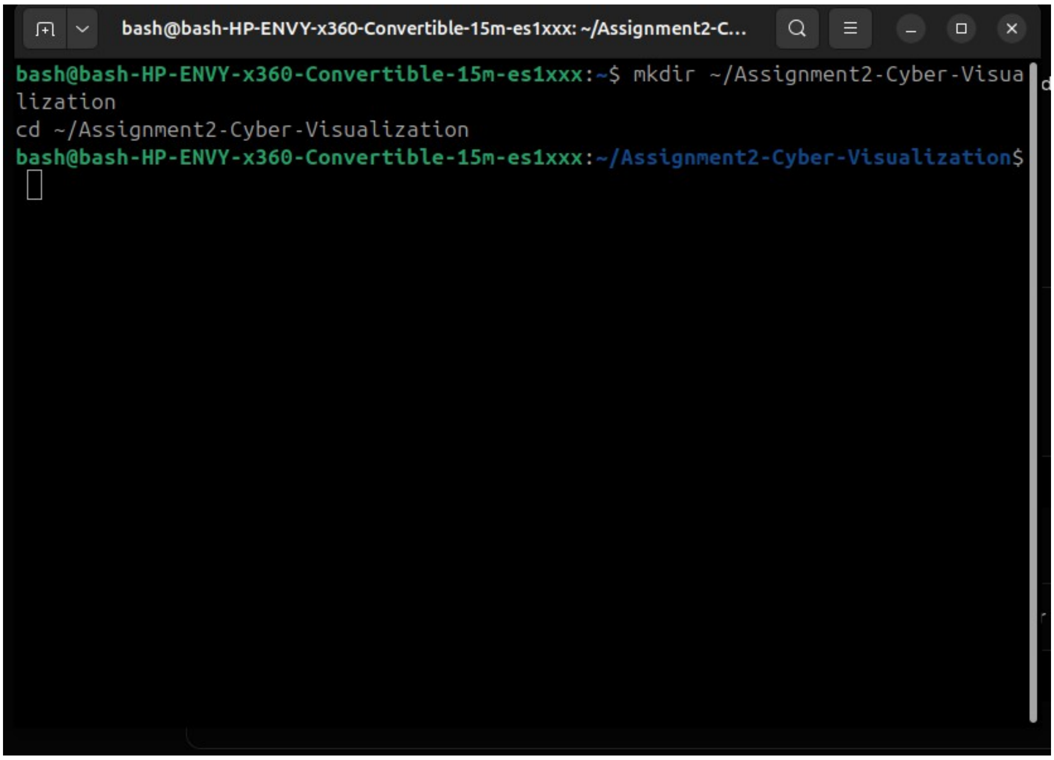


Figure 3.1 Creating the project folder in terminal

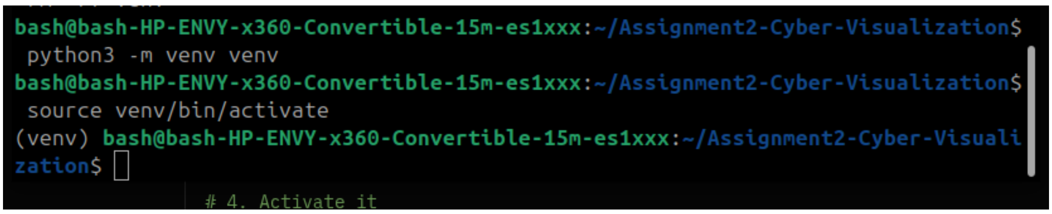


Figure 3.2 Creating and activating the virtual environment

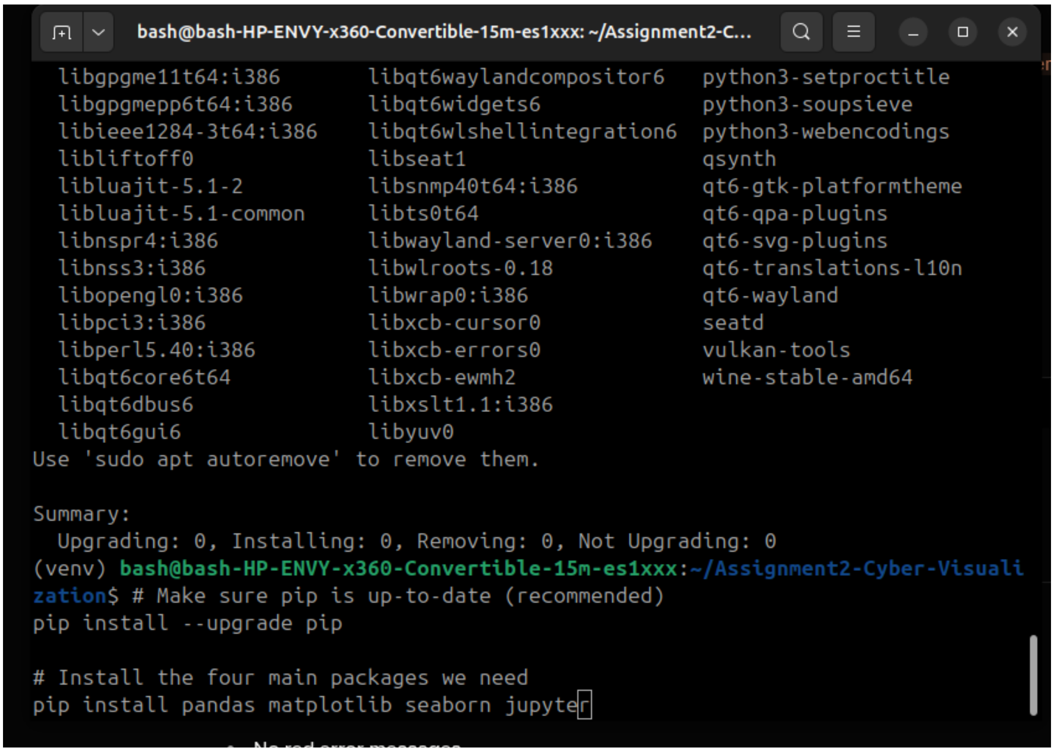


Figure 3.3 Installing required libraries (pandas, matplotlib, seaborn, jupyter)

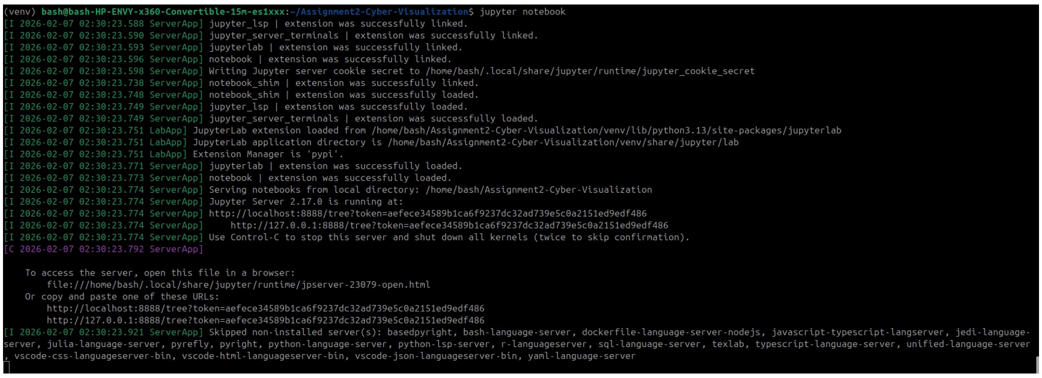


Figure 3.4 Launching Jupyter Notebook

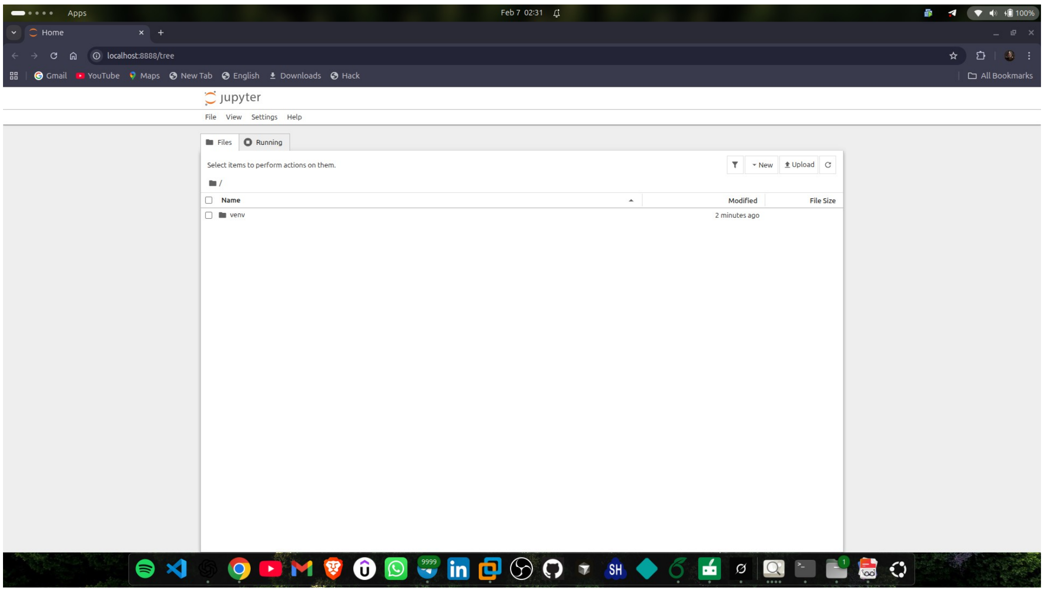


Figure 3.5 Jupyter Notebook home page showing project folder

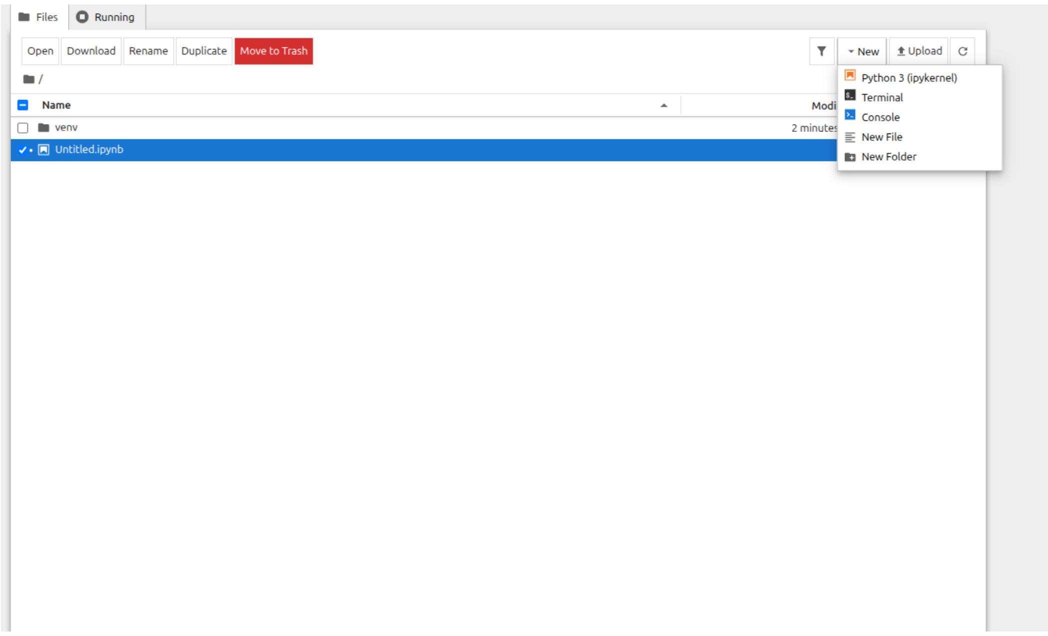


Figure 3.6 Creating a new notebook

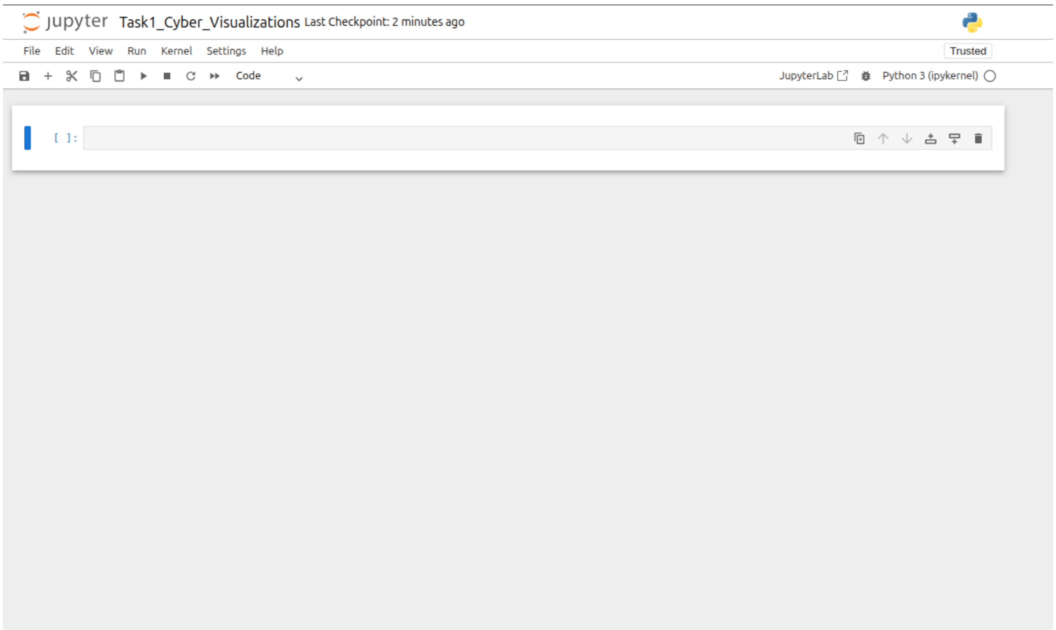


Figure 3.7 Notebook renamed to `Task1_Cyber_-Visualizations.ipynb`

3.1.2 Dataset Loading and Exploration

The dataset used is the public **Cybersecurity Attacks Dataset** from Kaggle.

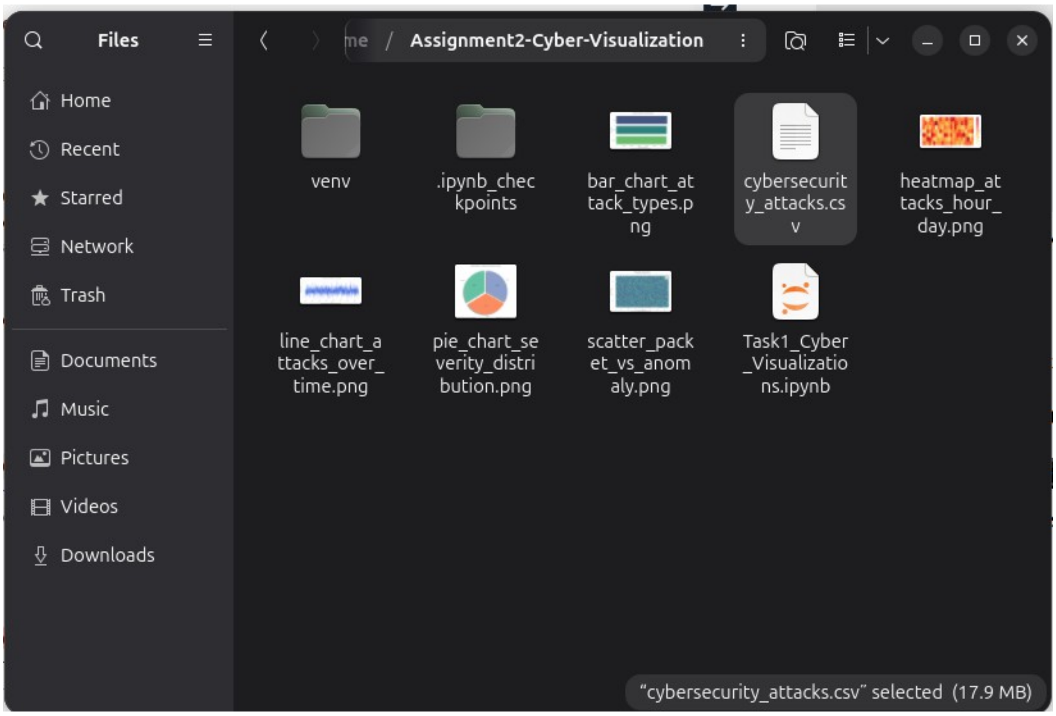


Figure 3.8 Dataset file `cybersecurity_attacks.csv` in the project folder

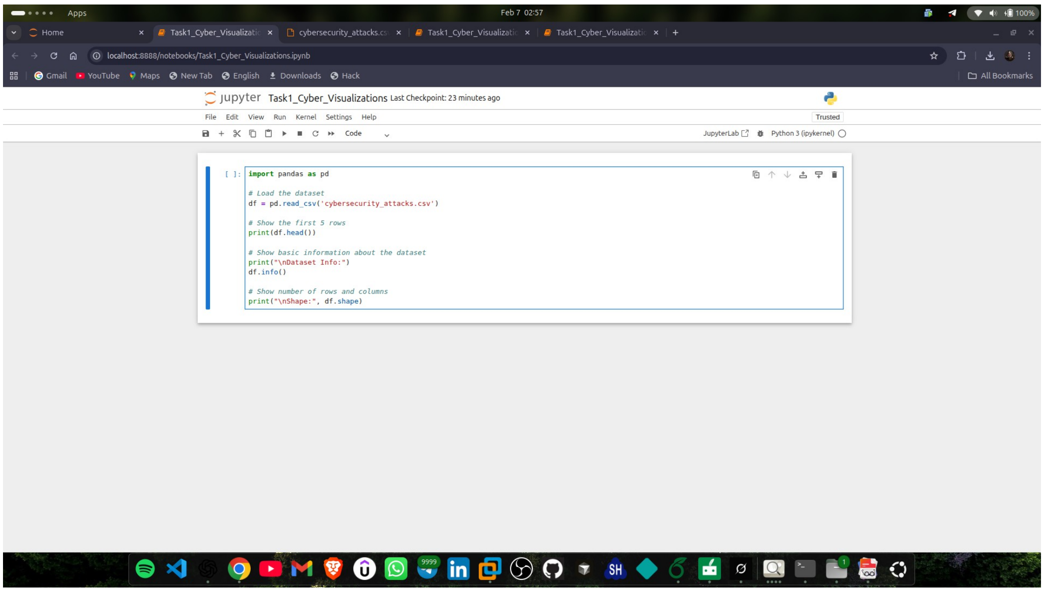


Figure 3.9 Code: Loading the dataset with pandas

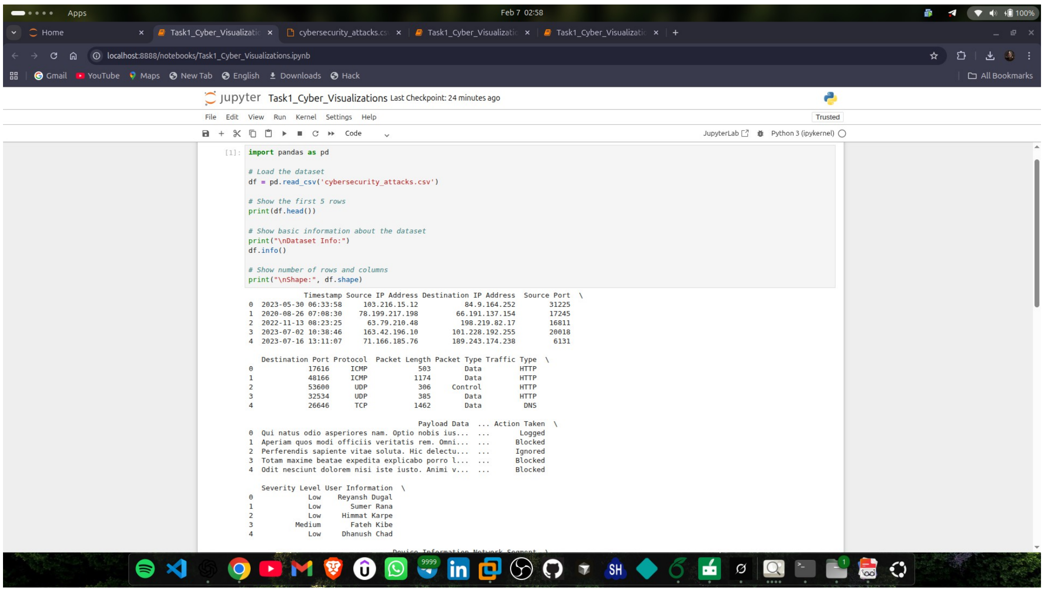


Figure 3.10 Output: Dataset information (df.info())

```
[2]: # Convert Timestamp to proper datetime (very important for time-based plots)
df['Timestamp'] = pd.to_datetime(df['Timestamp'])

# Quick check: show the first few timestamps after conversion
print("First few timestamps after conversion:")
print(df['Timestamp'].head())

# Show list of unique attack types and their counts
print("\nAttack Types and counts:")
print(df['Attack Type'].value_counts())

# Optional: total number of unique attack types
print("\nNumber of unique attack types:", df['Attack Type'].nunique())

First few timestamps after conversion:
0    2023-05-30 06:33:58
1    2020-08-26 07:08:30
2    2022-11-13 08:23:25
3    2023-07-02 10:38:46
4    2023-07-16 13:11:07
Name: Timestamp, dtype: datetime64[us]

Attack Types and counts:
Attack Type
DDoS         13428
Malware      13307
Intrusion    13265
Name: count, dtype: int64

Number of unique attack types: 3

[ ]:
```

Figure 3.11 Converting Timestamp to datetime and checking attack type distribution

3.1.3 Visualizations Created

Visualization 1 – Bar Chart

Title: Number of Attacks by Attack Type **Purpose:** Compare frequency of different attack types

```
[ ]:

[ ]: import matplotlib.pyplot as plt
import seaborn as sns

# Set a nice style for all plots (optional but looks better)
sns.set_style("whitegrid")

# Create the bar chart
plt.figure(figsize=(10, 6))

# Use seaborn countplot (very convenient for categorical counts)
sns.countplot(
    data=df,
    y='Attack Type',
    order=df['Attack Type'].value_counts().index,
    palette='viridis'
)

plt.title('Number of Attacks by Attack Type', fontsize=14, pad=15)
plt.xlabel('Number of Attacks', fontsize=12)
plt.ylabel('Attack Type', fontsize=12)

# Make sure long labels are readable
plt.tight_layout()

# Save the plot as an image file (you'll need this for the report)
plt.savefig('bar_chart_attack_types.png', dpi=150, bbox_inches='tight')

# Show the plot in the notebook
plt.show()
```

Figure 3.12 Code used to generate the bar chart

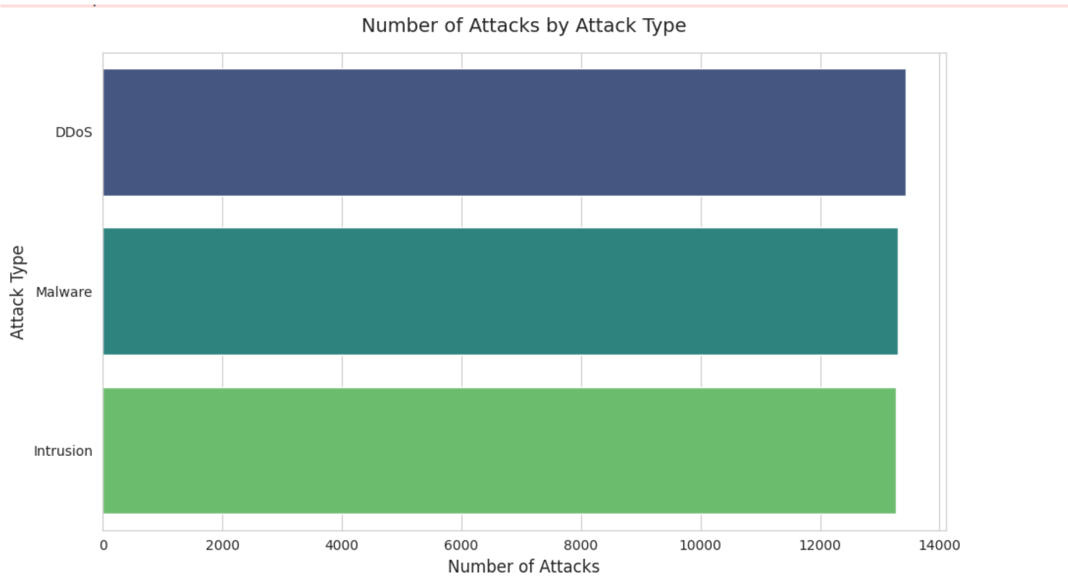


Figure 3.13 Bar chart – Number of Attacks by Attack Type

Visualization 2 – Line Chart

Title: Number of Attacks Over Time **Purpose:** Show temporal trends in attack activity

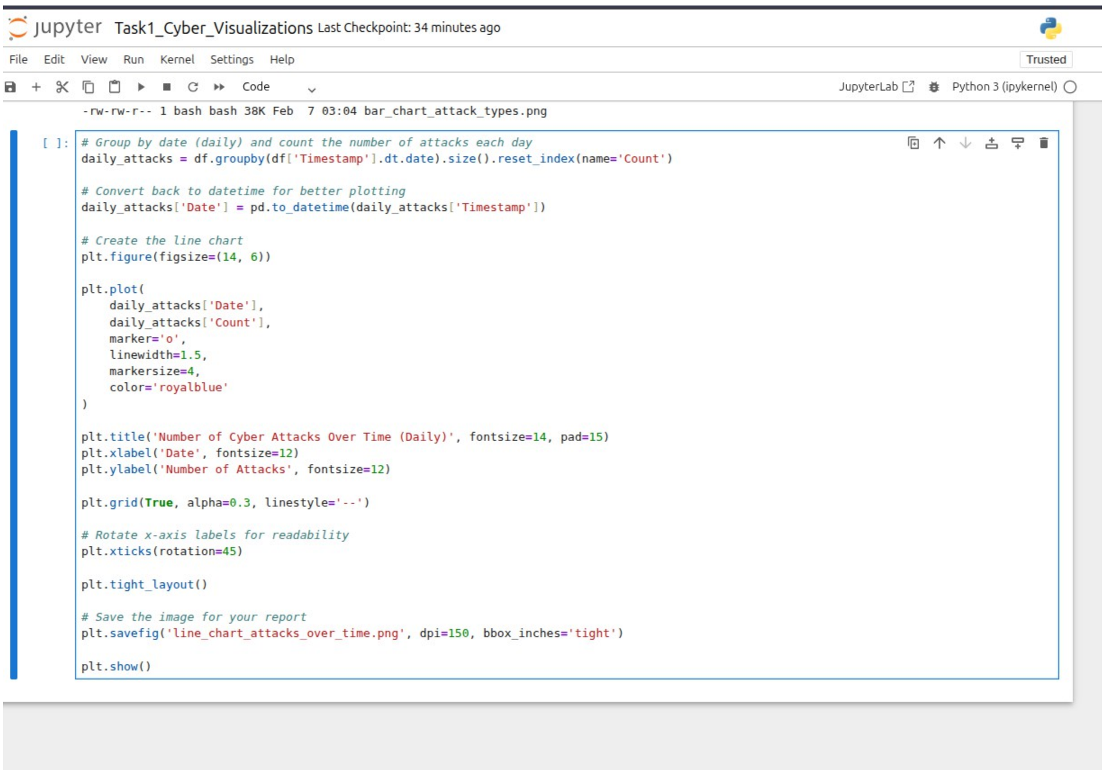


Figure 3.14 Code used to generate the line chart

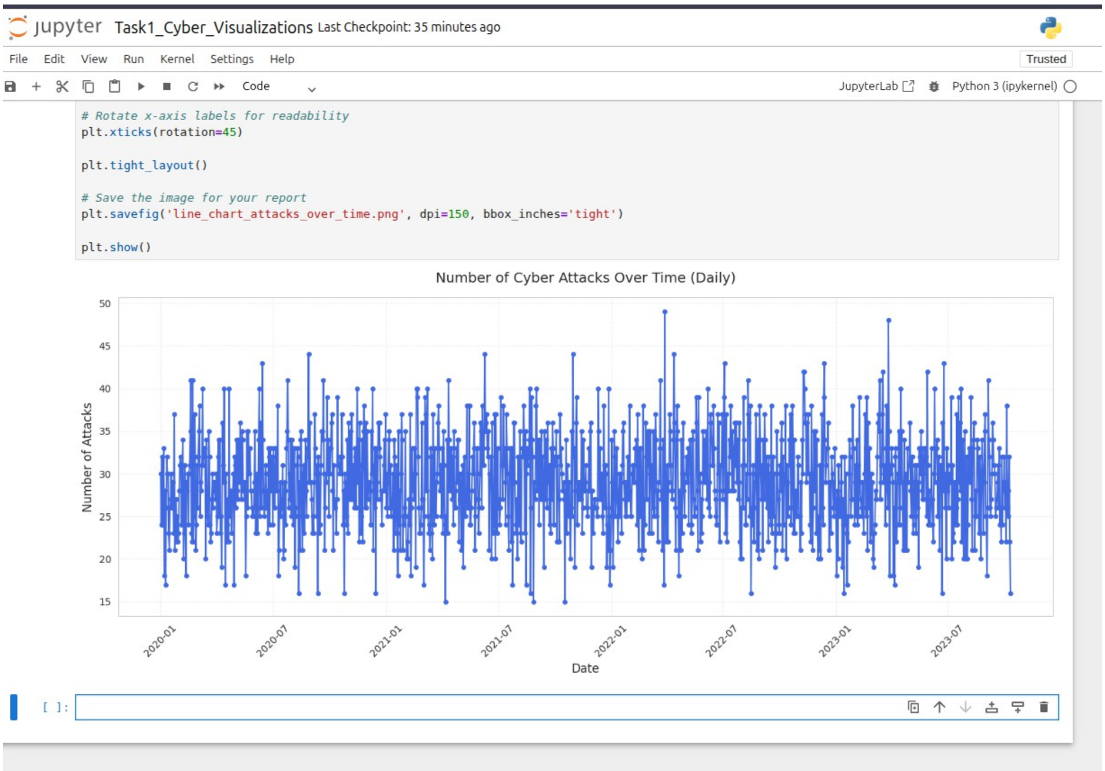


Figure 3.15 Line chart – Attacks over time (daily)

Visualization 3 – Heatmap

Title: Attack Frequency by Hour and Day of Week **Purpose:** Identify temporal patterns (time of day vs day of week)

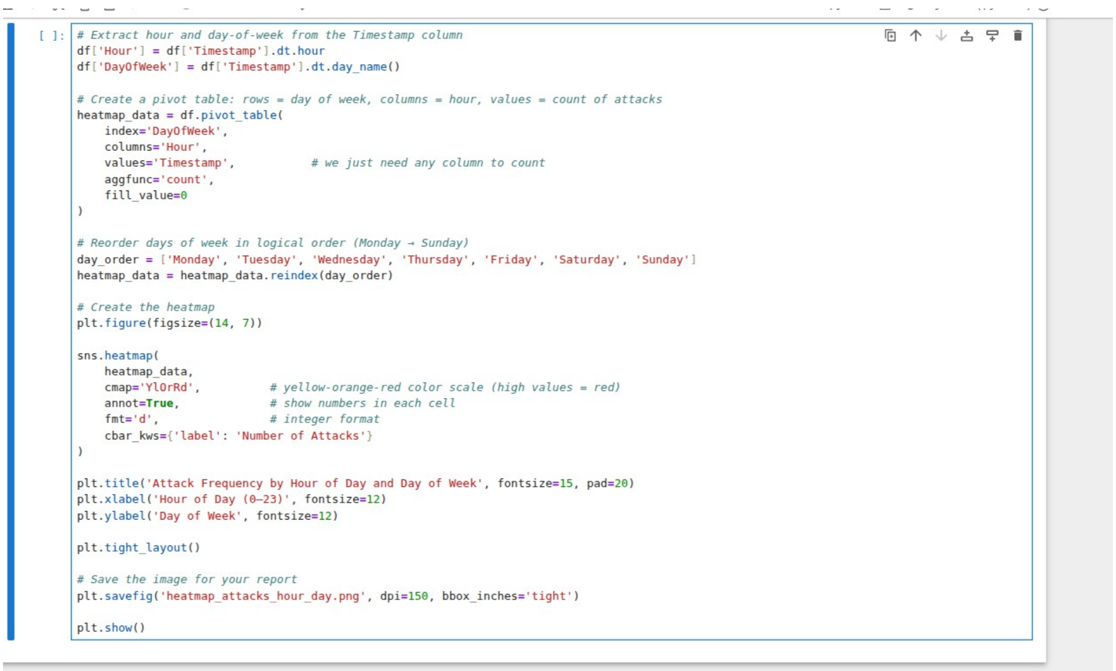
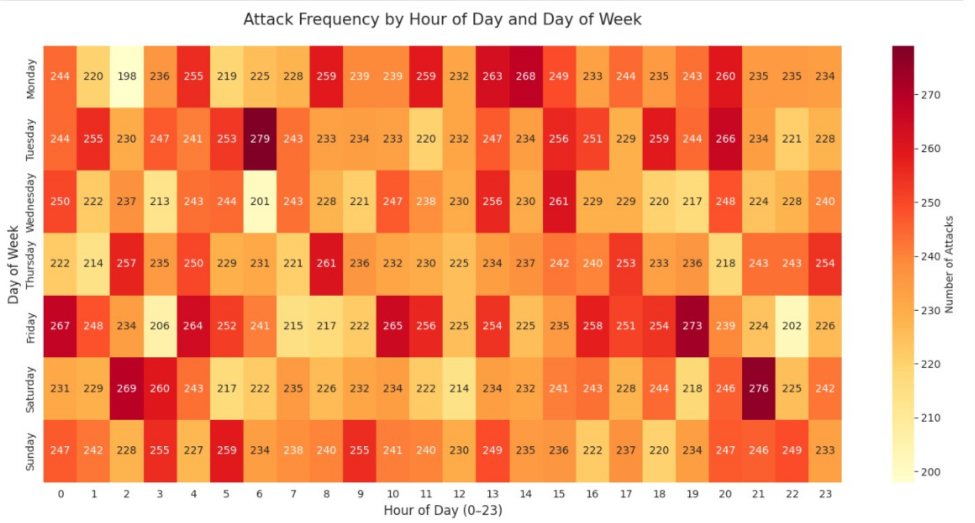


Figure 3.16 Code used to generate the heatmap




```
[9]: !ls -lh *.png
-rw-rw-r-- 1 bash bash  38K Feb  7 03:04 bar_chart_attack_types.png
-rw-rw-r-- 1 bash bash 219K Feb  7 03:12 heatmap_attacks_hour_day.png
-rw-rw-r-- 1 bash bash 275K Feb  7 03:08 line_chart_attacks_over_time.png
[ ]:
```

Figure 3.17 Heatmap – Attack frequency by hour and day

Visualization 4 – Pie Chart (additional)

Title: Distribution of Attack Severity Levels **Purpose:** Show the proportion of attacks by severity (Low, Medium, High, Critical)

```
[ ]: import matplotlib.pyplot as plt
import seaborn as sns

# Optional: set style (consistent with previous plots)
sns.set_style("whitegrid")

# Get the counts of each severity level
severity_counts = df['Severity Level'].value_counts()

# Create the pie chart
plt.figure(figsize=(10, 8))

plt.pie(
    severity_counts,
    labels=severity_counts.index,
    autopct='%1.1f%%',           # show percentage with 1 decimal place
    startangle=90,                # start from top
    colors=sns.color_palette('Set2', len(severity_counts)),
    shadow=True,
    explode=[0.05] * len(severity_counts)   # small separation between slices
)

plt.title('Distribution of Attack Severity Levels', fontsize=15, pad=20)

# Equal aspect ratio ensures the pie is drawn as a circle
plt.axis('equal')

# Save high-quality image for report
plt.savefig('pie_chart_severity_distribution.png', dpi=150, bbox_inches='tight')

plt.show()
```

Figure 3.18 Code used to generate the pie chart

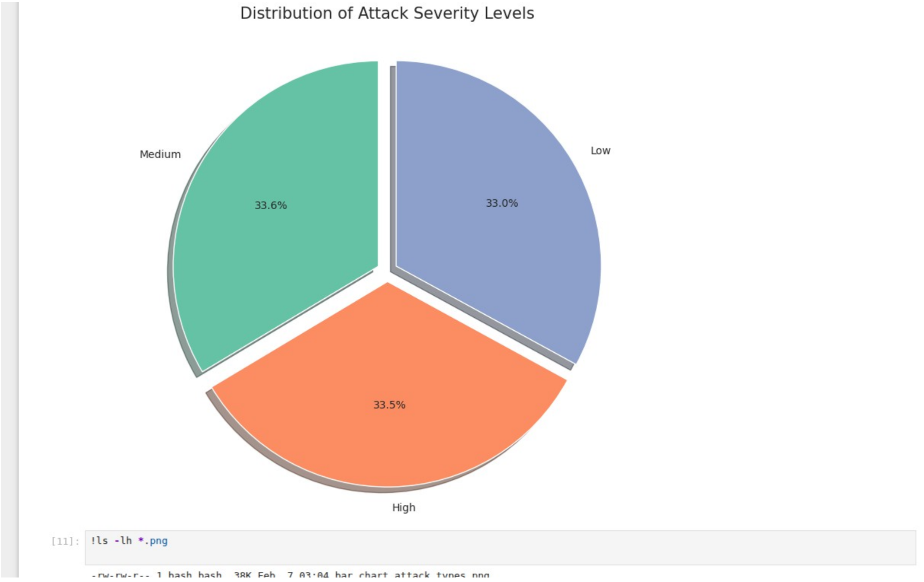


Figure 3.19 Pie chart – Distribution of attack severity levels

Visualization 5 – Scatter Plot (additional)

Title: Packet Length vs Anomaly Score **Purpose:** Explore relationship between packet size and anomaly score (colored by severity)

```
-rw-rw-r-- 1 bash bash  38K Feb  7 03:04 bar_chart_attack_types.png
-rw-rw-r-- 1 bash bash 219K Feb  7 03:12 heatmap_attacks_hour_day.png
-rw-rw-r-- 1 bash bash 275K Feb  7 03:08 line_chart_attacks_over_time.png
-rw-rw-r-- 1 bash bash  94K Feb  7 03:14 pie_chart_severity_distribution.png
```

```
[ ]: import matplotlib.pyplot as plt
import seaborn as sns

# Create the scatter plot
plt.figure(figsize=(12, 8))

sns.scatterplot(
    data=df,
    x='Packet Length',
    y='Anomaly Scores',
    hue='Severity Level',
    size='Severity Level',
    palette='viridis',
    alpha=0.7,
    sizes=(20, 120)           # larger points for higher severity
)

plt.title('Packet Length vs Anomaly Scores\n(colored by Severity Level)',
          fontsize=14, pad=15)

plt.xlabel('Packet Length (bytes)', fontsize=12)
plt.ylabel('Anomaly Score', fontsize=12)

# Add a grid for better readability
plt.grid(True, alpha=0.3, linestyle='--')

plt.tight_layout()

# Save high-resolution image for the report
plt.savefig('scatter_packet_vs_anomaly.png', dpi=150, bbox_inches='tight')

plt.show()
```

Figure 3.20 Code used to generate the scatter plot

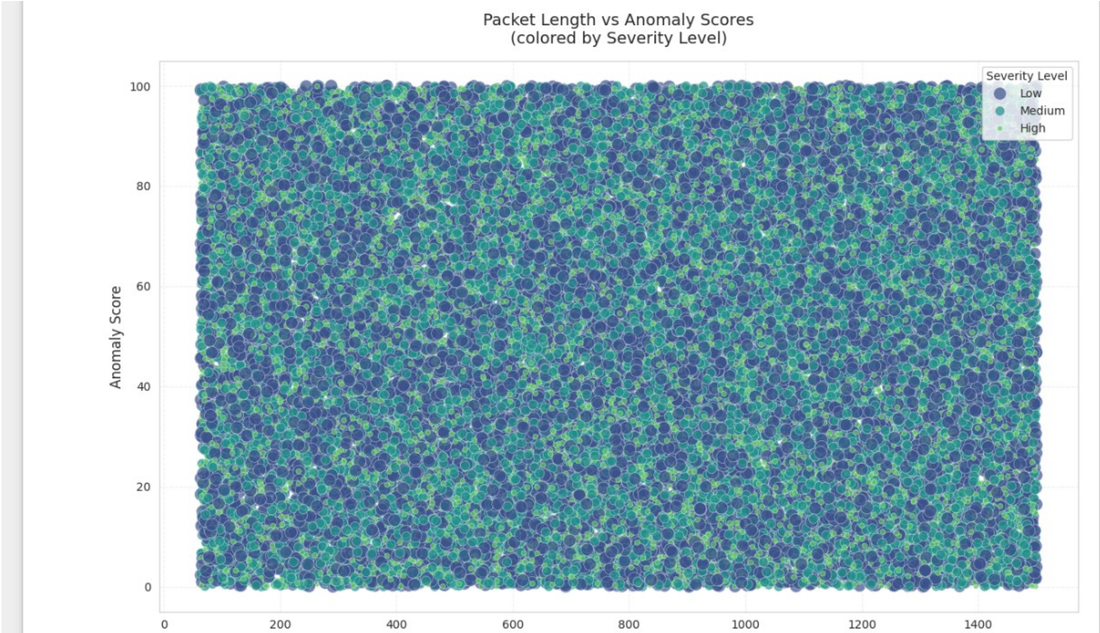


Figure 3.21 Scatter plot – Packet length vs anomaly score (colored by severity)

CHAPTER 4

SCREENSHOTS OF CYBER-SPECIFIC VISUALIZATION TOOL

This chapter presents the screenshots demonstrating the installation, configuration, data import, and visualization processes performed using the selected cyber-specific visualization tool. The figures provide visual evidence of the practical implementation described in Chapter Three.

4.1 Tool Installation

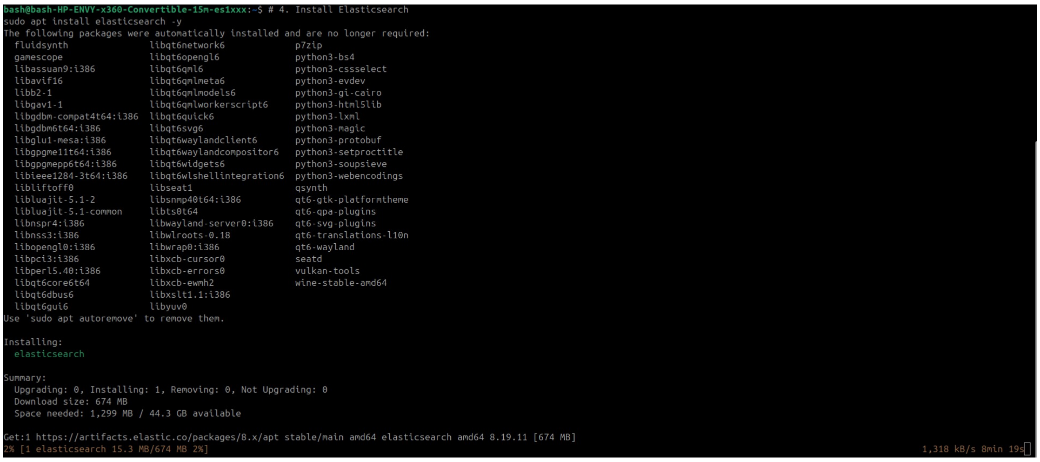


Figure 4.1 Download Page of the Cyber-Specific Visualization Tool(Elastic Search)

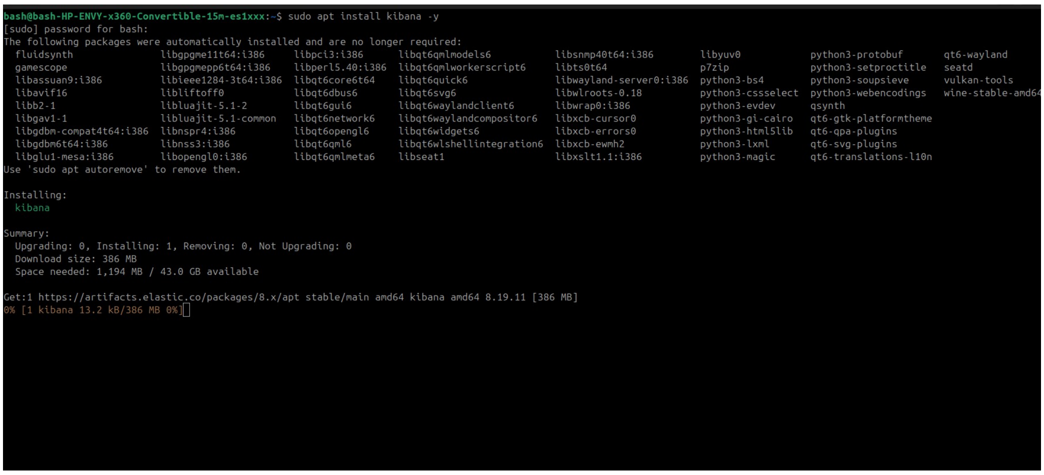


Figure 4.2 Installation of the Cyber-Specific Tool(Kebana)

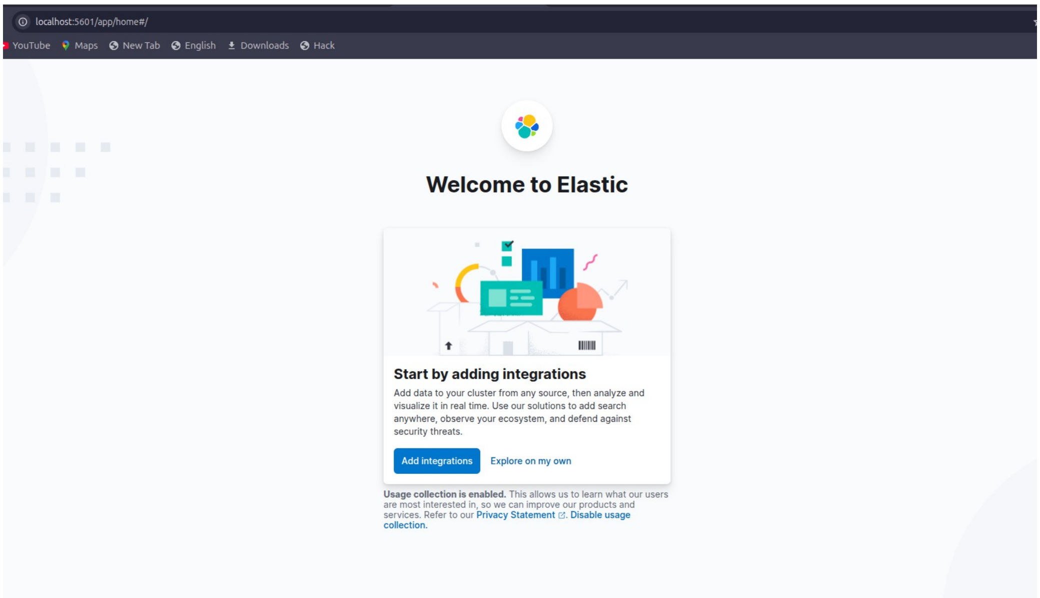


Figure 4.3 Successful Launch of the Cyber-Specific Visualization Tool

4.2 Dataset Import and Configuration

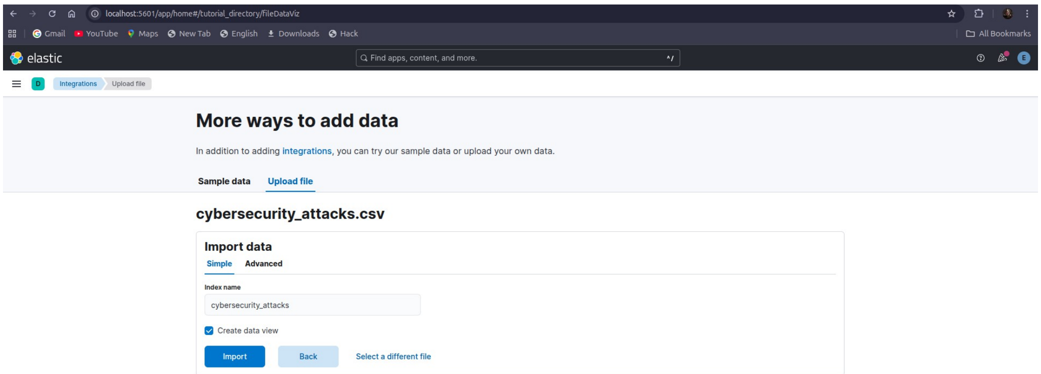


Figure 4.4 Dataset Upload Interface

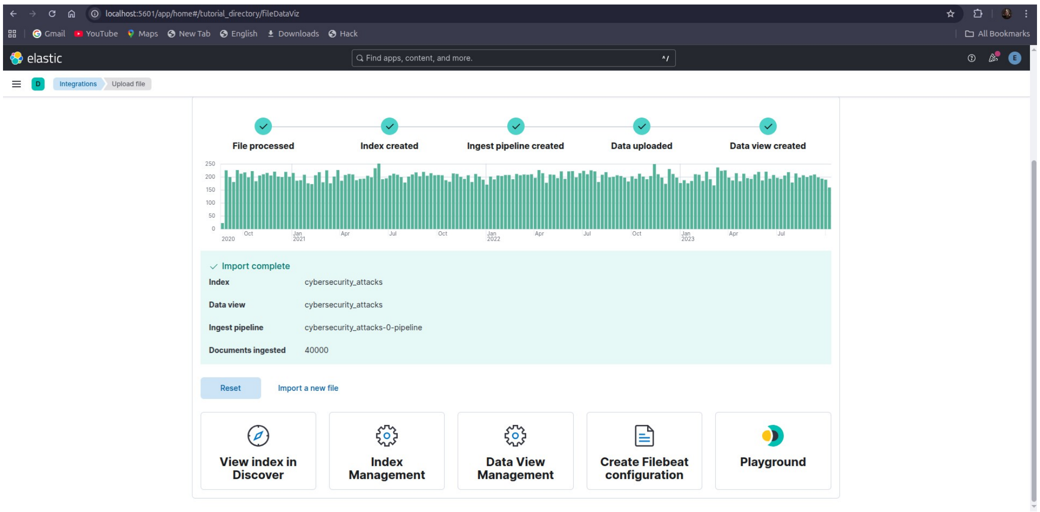


Figure 4.5 Confirmation of Successful Dataset Import/Indexing

4.3 Visualization 1: Timeline of Security Events

This visualization presents a timeline of recorded cybersecurity events in the dataset. By aggregating attack occurrences over time, it allows the identification of trends, spikes, and periods of increased malicious activity. Such a timeline is useful for detecting coordinated attacks and understanding temporal behavior of security incidents.

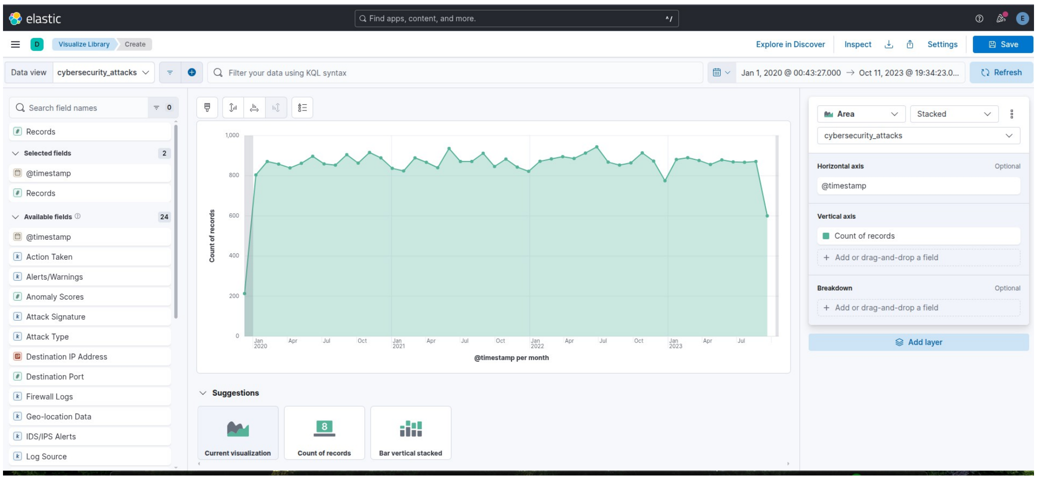


Figure 4.6 Timeline of Security Events Showing Attack Frequency Over Time

4.4 Visualization 2: Attack Pattern Heatmap

The heatmap visualization illustrates attack intensity based on time and attack type. Darker color regions represent periods with higher attack frequency, revealing recurring patterns and peak activity windows. This visualization supports proactive threat detection by highlighting predictable attacker behavior.

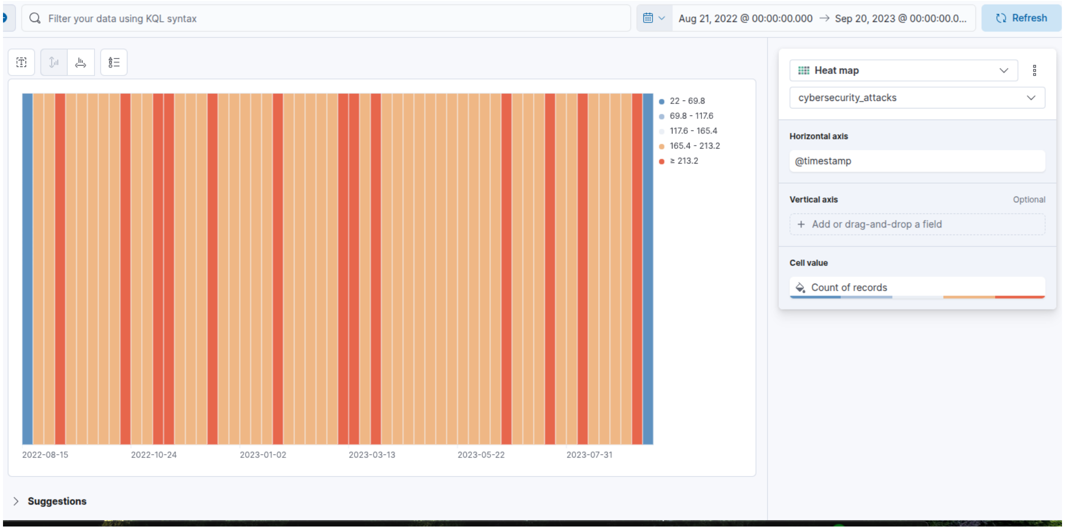


Figure 4.7 Heatmap Showing Attack Patterns by Time and Attack Type

4.5 Visualization 3: Top 10 Attack Types

This bar chart displays the top ten most frequent attack types observed in the dataset. The visualization clearly indicates that Distributed Denial of Service (DDoS) attacks occur more frequently than other attack categories, emphasizing the need to prioritize availability protection and DDoS mitigation strategies.

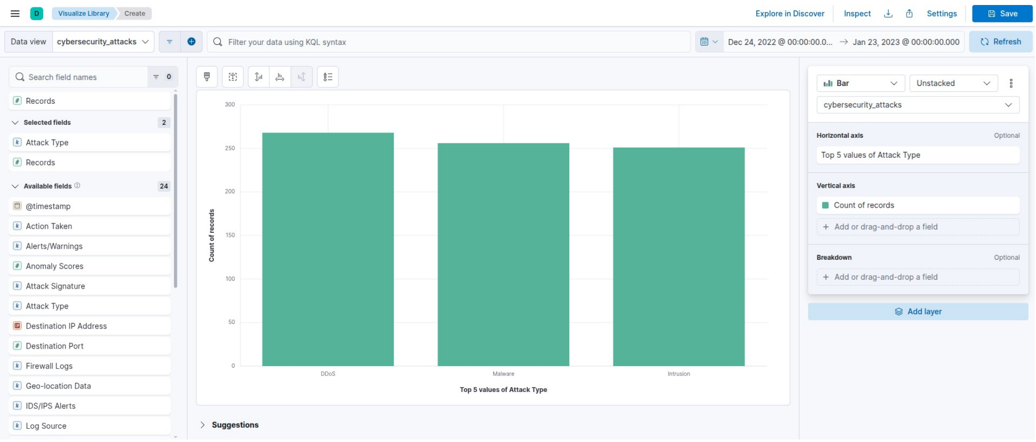


Figure 4.8 Top 10 Attack Types Highlighting the Prevalence of DDoS Attacks

4.6 Conclusion

This project demonstrated the practical application of data visualization techniques in the analysis of cybersecurity datasets. By employing both general-purpose and cyber-specific visualization tools, meaningful insights were extracted from complex security data. The visualizations developed provided clear perspectives on attack trends, patterns, and dominant threat types.

The cyber-specific analysis using Kibana revealed continuous attack activity over time, with noticeable variations in intensity across different periods. The timeline visualization highlighted temporal trends, while the heatmap exposed recurring attack patterns and peak activity windows. Furthermore, the top attack types visualization identified Distributed Denial of Service (DDoS) attacks as the most frequent threat in the dataset, underscoring their significant impact on system availability.

Overall, this project highlights the importance of visualization in cybersecurity monitoring and decision-making. Effective visual representations enable security analysts to quickly interpret large volumes of data, prioritize threats, and support proactive defense strategies. The skills and techniques demonstrated in this work are essential for modern cybersecurity operations and incident response.